Angle Rules

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS acute · obtuse · reflex · straight line · angles at a point · vertically opposite

A Types of angles ●○○ ANSWER ON THE SHEET

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 35° _____ **b)** 90° _____ **c)** 120° _____ **d)** 180° _____

e) 270° _____ **f)** 89° _____ **g)** 91° _____ **h)** 300° _____

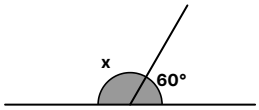
i) 12° _____ **j)** 179° _____ **k)** 181° _____ **l)** 95° _____

B Angles on a straight line ●○○ ANSWER ON THE SHEET

Remember: Angles on a straight line add to 180°.

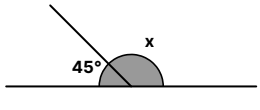
2. Find the size of the angle marked x. Diagrams are not drawn accurately.

a)



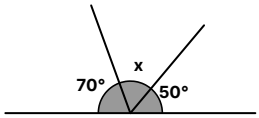
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b)



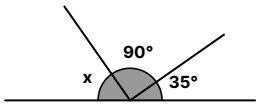
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c)



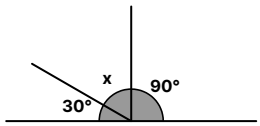
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d)



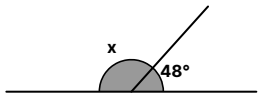
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e)



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f)



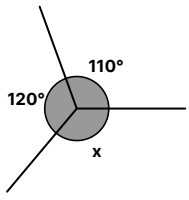
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C Angles at a point ●●○ ANSWER ON THE SHEET

Remember: Angles that go all the way around a point add to 360°.

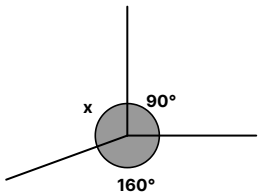
3. Find the size of the angle marked x.

a)



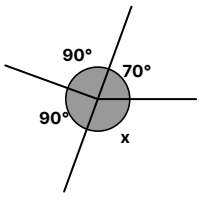
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b)



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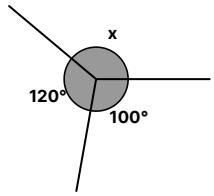
c)



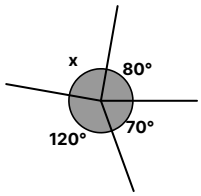
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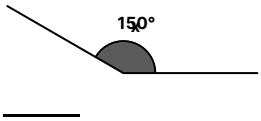
d)



e)



f)



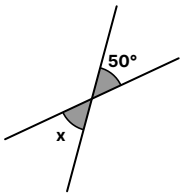
D Vertically opposite angles ●●○

ANSWER ON THE SHEET

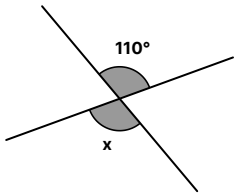
Remember: When two straight lines cross, the angles opposite each other are equal.

4. Find the size of the angle marked x.

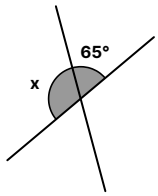
a)



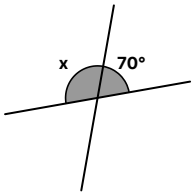
b)



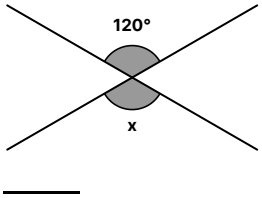
c)



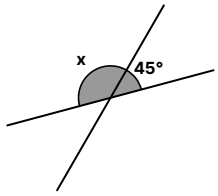
d)



e)



f)



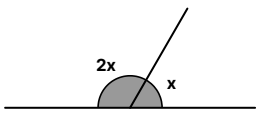
E Using algebra ●●●

WORK IN YOUR BOOK

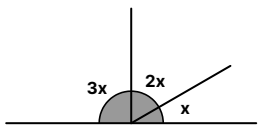
Remember: Write an equation using the angle rule, then solve it.

5. Write an equation and solve it to find x.

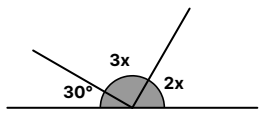
a)



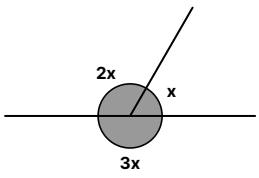
b)



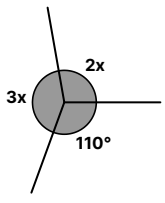
c)



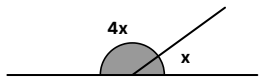
d)



e)



f)



★ Extension

FOR EARLY FINISHERS

6. Two straight lines cross. One angle is $(2x + 10)^\circ$ and the angle vertically opposite it is 70°. Find x.

7. Three angles lie on a straight line. One is 55° and the other two are equal. Find the size of each of the equal angles.

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Cell Tuition
1-1 PRIVATE TUTORS

YEAR 9 · GEOMETRY

Angle Rules

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS acute · obtuse · reflex · straight line · angles at a point · vertically opposite

A Types of angles ●○○ ANSWER ON THE SHEET

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 50° _____ **b)** 90° _____ **c)** 140° _____ **d)** 180° _____

e) 250° _____ **f)** 88° _____ **g)** 92° _____ **h)** 320° _____

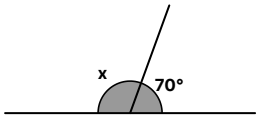
i) 7° _____ **j)** 178° _____ **k)** 182° _____ **l)** 100° _____

B Angles on a straight line ●○○ ANSWER ON THE SHEET

Remember: Angles on a straight line add to 180°.

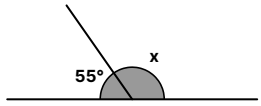
2. Find the size of the angle marked x. Diagrams are not drawn accurately.

a)



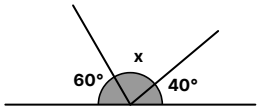
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b)



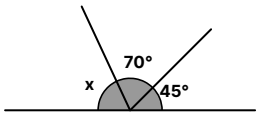
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c)



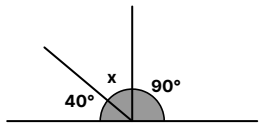
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d)



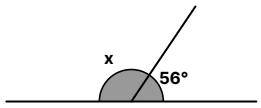
_____

e)



_____

f)



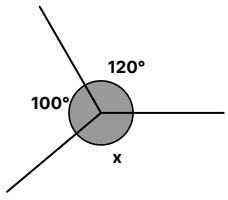
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C Angles at a point ●●○ ANSWER ON THE SHEET

Remember: Angles that go all the way around a point add to 360°.

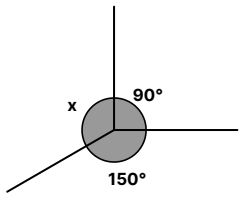
3. Find the size of the angle marked x.

a)



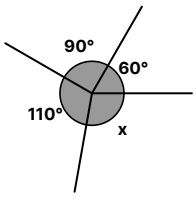
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b)



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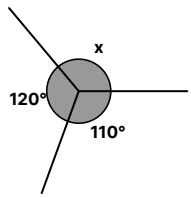
c)



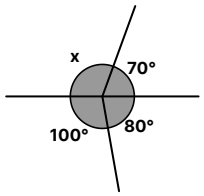
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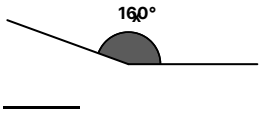
d)



e)



f)



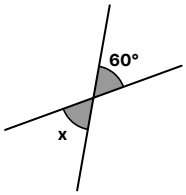
D Vertically opposite angles ●●○

ANSWER ON THE SHEET

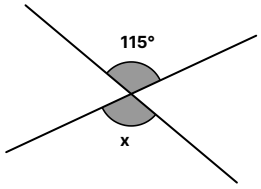
> **Remember:** When two straight lines cross, the angles opposite each other are equal.

4. Find the size of the angle marked x.

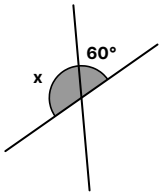
a)



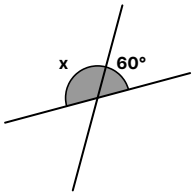
b)



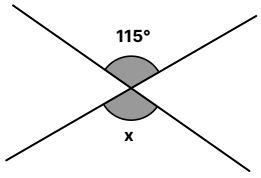
c)



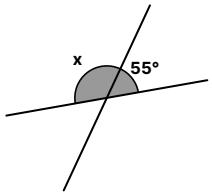
d)



e)



f)



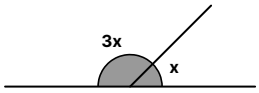
E Using algebra ●●●

WORK IN YOUR BOOK

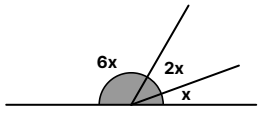
> **Remember:** Write an equation using the angle rule, then solve it.

5. Write an equation and solve it to find x.

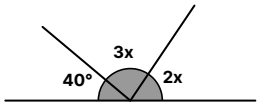
a)



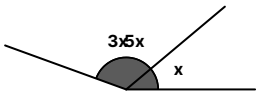
b)



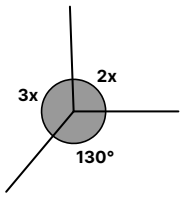
c)



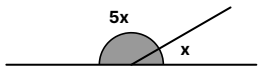
d)



e)



f)



★ Extension

FOR EARLY FINISHERS

6. Two straight lines cross. One angle is $(3x + 15)^\circ$ and the angle vertically opposite it is 75°. Find x.

7. Three angles lie on a straight line. One is 65° and the other two are equal. Find the size of each of the equal angles.

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